

1 MPL-TRA - Create Transport Order

Overview

License	MPL-TRA
---------	---------

Purpose

Use this function to create a transport order for a planned material quantity, batch or resource.

Transport orders are created, for example

- when output batches are generated (dialog: CA_WL, CA_AB, A_UN, A_AB)
- when operations are planned via the Shop Floor Scheduling module (HLS.SAVE)
- manually via a client function

Requirements

The database patch **dbp_mpl_transportation.hsc** must have been run.



Configure an active work plan with corresponding order type to create the transport order.

Features

Input parameters:

Order type (TRANR.AUART)

You need the order type to create the transport order from a work plan. The work plan must be stored and activated in the work plan management.

For the batch-related transport, the system uses the article number to search for a work plan. If the article number cannot be found, the system searches for a work plan without article number.

The system uses a work plan that does not have an article number to transport resources.

Final article (TRANR.ATK)

The final article number specifies the type of material that is to be transported. This number is transferred to the transport order. If a batch is transported, the entered number must match the article of the batch. If a resource is transported, the article number is irrelevant.

It is a mandatory field if batches are transported.

Batch number (TRANR.CNR)

You have to enter an existing and free batch with batch class (G) "yield" to transport batches. Once generated successfully, the batch is assigned to the transport order and added the transport status B (stands for "transport prepared"). The batch must be included in the source material buffer.

Resource type (TRANR.RESTYP)

You have to enter the resource type to transport a resource.

Resource (TRANR.RES)

You have to enter the resource number and resource type for a unique assignment if you want to transport a resource. You can only transport a resource if it is not yet assigned to a transport order and if it is not in the status "locked". For this purpose, the resource must be included in the "preceding material buffer".

Preceding material buffer (TRANR.SMP)

If you want to transport batches, the batch must be included in the preceding material buffer. This is the material buffer where the transport starts.

Subsequent material buffer (TRANR.TMP)

This material buffer is the target buffer where the batch or resource is to be transported to.

Start date (TRANR.DATFB)

The date is entered in the *Earliest start date* field of the transport order.

End date (TRANR.DATSE)

The date is entered in the *Latest end date* field of the transport order.

Synchronous or asynchronous processing (TRANR.ASYNC)

TRANR.ASYNC=J: Entry in queue table `mpl_tranr_queue`. A scheduler job (*tranr_screate.scr*) processes the entries at regular intervals. The bapi **TRANR.SCREATE** processes the queue table data.

The following checks are made if you call the bapi TRANR.CREATE.

- The order type must exist.
- An active work plan must exist for the order type.
- The resource or batch must be included in the source material buffer of the transport order.
- Checks for batch-related transport:
 - The batch must exist.
 - The batch must be free (status = F and quality status unequal to S).

- Batches that are in the *transport* status must not be transported again.
- Checks for article-related transport
 - The calculated required quantity in relation to the input quantity of the component and the target quantity of the planned operation must be > 0.
- Optionally, the staff badge number is checked (KNR) if entered.

Result of BAPI call TRANR.CREATE:

- The transport order has been created.
- The transport operation has been created with the OP status I (initial).
- The following status information (auftrag_status table) is stored with the transport order:
 - Planned production order (field trigger_anr)
 - Transport type (field tranr_art)
 - A = article-related quantity from planning
 - L = for batch
 - R = for resource
 - Object reference (fields tranr_cnr, tranr_res and tranr_restyp)
- Source material buffer (field tranr_smp)

Transport type A from planning:
The output buffer of the preceding OP specifies the source material buffer. The buffer remains empty, if the source material buffer cannot be identified via the preceding OP.

Transport type L for output batch change:
Output buffer of the current machine.
- Target material buffer (field tranr_tmp)

Transport type A from planning:
Input buffer of the current machine where the operation was planned.

Transport type L for output batch change:
Input buffer of the following machine of the subsequent operation.
- In case of batch-related objects, the created transport order includes the quantity of the batch in the "target quantity" field. The quantity 1 is set by default for the target quantity of resources. For article-related transports, the required quantity (based on the component's input quantity) is entered as the target quantity.
- For an article-related transport order (transport type = A) that was created due to planning, the batch will only be assigned upon starting.
- The object to be transported switches to the status
 - Batch:** Transport status = B (stands for "transport prepared")
Impact: As long as the batch is in the transport status B, it cannot be logged on as input batch.
 - Resource:** Status does not change.

Configuration

Default order type

The following HYD-INI option specifies the order type when creating transport orders from planning or when changing output batches:

Name=MPL
Section=TRANSPORTATION
Parameter=TRANR_AUART
Value=<Order type>

Work plan:

An active work plan with the corresponding order type is required to create transport orders. The work plan must be created with the order type and the corresponding article. The work plan must be activated via work plan management. There is still also the possibility to use a work plan without article number for all versions (e.g. for resources).

Further properties of the work plan/transport order:

The work plan includes exactly one operation that planned for the machine group to execute the transport.

The transport operation is not subject to batch management.

You cannot log on transport operations several times.

Escalation - TRANR.TAP_NOT_FOUND

The following escalation is triggered if no active work plan can be found when creating a transport order:

Parameter:
TRANR.ATK=<article number>
TRANR.AUART=<order type>
TRANR.SANR=<triggering OP>
TRANR.DAT=<current date>
TRANR.ZEI=<current time>

Escalation - TRANR.TO_ACTIV

The following escalation is triggered if no active work plan can be found when creating a transport order:

This escalation is triggered if a production operation is deallocated (cancelled/unplanned) that is already assigned to an active transport order.

Parameter:

TRANR.AUNR=<order number of transport order>

TRANR.ANR=<OP number of transport order>

TRANR.ANR_STA=<OP status of transport order>

TRANR.SANR=<deallocated/cancelled OP>

TRANR.DAT=<current date>

TRANR.ZEI=<current time>

Machine:

You can configure for the machine if a transport order is to be generated in relation to the input material when output batches are changed or upon planning.

Material type:

You can configure for the material type if a transport order is to be generated in relation to the input material when output batches are changed or upon planning. The material type configuration overrides the configuration of the machine.

Scheduler – Create transport orders from planning

Enter a cyclic job to call the bapi **TRANR.SCREATE** in the scheduler:

Command: sh.exe ./tranr_screate.scr

Comment: MPL-TRA: create transport order from planning

Interval: 10 minutes